MIL-C-50443 (MU)
22 September 1969
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

CARTRIDGE, 105MM, HE: M444E1,
BODY ASSEMBLY AND BASE PLUG FOR

1. SCOPE

1.1 This specification covers the body assembly and base plug for one type of Cartridge designated as Cartridge, 105MM, HE: M444E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-B-20292 - Blanks, Rotating Band, for Projectiles
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing
FED-STD-151 - Metals; Test Methods

FSC: 1315

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9245951 - Body Assembly
9245953 - Plug, Base
8837835 - Pallet for Fixed and Semi-Fixed Shells

PUBLICATIONS

ARMY

ET9245951 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E 18-67 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

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3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Body assembly and base plug.-The body assembly and base plug shall comply with all requirements on Drawings (Dwgs) 9245951 and 9245953, all associated drawings and with all requirements specified in applicable specifications.

3.3 Rotating band clearance.-The clearance between the band and the band seat in a radial direction shall not exceed 0.006 inch. In addition, the band shall make contact with the sidwalls of the band seat for a full 360 degrees and a minimum of 80 percent of the height of the band seat wall when tested as specified in 4.4.3.

3.4 Mechanical properties of rotating band.-The rotating band, after removal from the body assembly, shall have the following mechanical properties when tested as specified in 4.4.5.

Tensile strength - 30,000 pound per square inch (psi), minimum
Elongation - 15 percent minimum in 1 inch

3.5 Bend test of rotating band.-The rotating band, after removal from the body assembly, shall comply with bend test requirements specified in MIL-B-20292 when tested as specified in 4.4.4.

3.6 Proving ground ballistic tests (first article sample only).-A first article sample (see 4.2.1 (h)), shall be subjected to the proving ground ballistic tests specified in 4.2.2.1 and shall show no evidence of metal parts failure when tested as specified in 4.4.9. (See 6.1) The following conditions shall be considered as evidence of failure:

(1) Loss, separation, break-up, or fracture of metal parts in gun bore or during flight.

(2) Shear or loss of rotating band.

(3) Metal parts failure during expulsion phase.

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3.7 Use of steel blooms or billets with hot-sheared ends (applicable when this process is used).-The contractor shall demonstrate his ability to produce acceptable projectiles, manufactured from semi-finished, forging quality, carbon steel blooms or billets with hot-sheared ends. This shall be demonstrated by satisfactorily producing the first article quantities (as evidenced by successful completion of all first article examinations and tests) and satisfactorily producing approximately six-thousand (6000) projectiles as first deliveries under the contract from the above stated material.

3.8 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Discoloration, not due to corrosive action, will be permitted. All required marking shall be neat and sharply defined.

3.8.1 Heat treatment or stress-relief.-The heat treatment or stress-relief applied shall be uniform throughout each lot of components, and shall be so performed and controlled as to produced in the components the mechanical properties specified on the applicable component drawing.

3.9 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.4)

3.10 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

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a. A statement that the lot complies with all the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authorized to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of the following:

a. Ten (10) unpainted base plugs with protective coating.
b. Five (5) painted base plugs.
c. Ten (10) unbanded, unpainted bodies with protective coating.
d. Ten (10) unpainted body assemblies with protective coating.
e. Five (5) painted body assemblies.
f. Twenty-five (25) rotating bands.
g. Three (3) slugs representing the steel blooms or billets with hot-sheared ends which were used in producing the body assemblies. (Applicable when this process is used).
h. Twenty (20) painted body assemblies and twenty (20) painted base plugs (see 4.2.2.1).

All parts and completed assemblies submitted shall have been produced by the contractor, or furnished by a supplier, using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance for materials. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever a change occurs in manufacturing process, material used, drawing, specification or source

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of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawing. In addition, the following tests shall be performed:

4.2.2.1 Proving ground ballistic tests (see 3.6).-The twenty (20) painted body assemblies and base plugs from 4.2.1 (h), shall be forwarded to the location designated by the Contracting Officer for loading. The loaded assemblies shall then be subjected to the ballistic tests listed below. Testing shall be conducted at a Government owned proving ground using the methods and procedures specified in 4.4.9.

4.2.2.1.1 Five (5) of the loaded assemblies will be fired at Zone 7. (See 4.4.9.1)

4.2.2.1.2 Five (5) of the loaded assemblies will be subjected to a five-foot drop test and fired at Zone 7. (See 4.4.9.2)

4.2.2.1.3 Ten (10) of the loaded assemblies will be fired at Zone 7X. (See 4.4.9.3)

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted (including successful completion of all ballistic tests), the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection Provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements.

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The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications, and same specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105.

Unless otherwise approved by the Contracting Officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period, nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain rotating bands of one lot interfix number from one manufacturer.

4.3.1.1 Heat treatment or stress-relief lot.-A heat treatment or stress-relief lot shall consist of bodies from one mill-heat of steel that have been heat-treated or stress-relived in the same furnace at the same temperature and for the same length of time. Test samples shall be selected after final heat treatment or stress-relief. Where heat treatment or stress-relief is done in a continuously operated, conveyor-type furnace, the identification maintained shall be that code number assigned the particular mill-heat of steel. Where heat treatment or stress-relief is performed in batch type furnaces, additional identification control must be maintained that directly relates to that portion of the mill-heat processed as a batch. The method of control must meet with the approval of the Government. Components in heat treatment or stress-relief lots shall remain grouped together for identifying purposes until mechanical properties tests have been completed. Components from heat-treatment or stress- relief lots which comply with the applicable requirements may be subsequently re-grouped into projectile lots.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105), except that, inspection for Critical defects shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government, and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each Major

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defect and an AQL of 0.65 percent for each Minor defect except where 100% inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Plug, base prior to painting (see dwg. 9245953)

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Press-fit diameter	Gage	01001
102.	Perpendicularity of press-fit diameter with body seat shoulder	Gage	01002
103.	Length to body seat shoulder	Gage	01003
104.	Largest diameter	Gage	01004
105.	Concentricity of largest diameter with press-fit diameter	Gage	01005
106.	Length to largest basic diameter	Gage	01006
107.	Concentricity of largest basic diameter with press-fit diameter including perpendicularity with body seat shoulder	Gage	01007
108.	Width of relief behind press-fit diameter	Gage	01008
109.	Finish improper on press-fit diameter	Visual	01009
Minor:	AQL 2.50 Percent		
201.	Total length	Gage	01010
202.	Diameter of relief behind press-fit diameter	Gage	01011
203.	Chamfer on press-fit diameter	Visual	01012
204.	Radius missing	Visual	01013
205.	Finish improper (excluding press-fit diameter)	Visual	01014
206.	Evidence of poor workmanship (see 3.8)	Visual	01015

4.3.2.2 Plug, base (see dwg. 9245953)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		

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Minor:	AQL 1.00 Percent		
201.	Protective coating inadequate (base metal exposed)	Visual	02001
202.	Evidence of poor workmanship (see 3.8)	Visual	02002

4.3.2.3 Body (see dwg. 9245952)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 2.50 Percent		
101.	Diameter of bourrelet, min (2 places)	Gage	03001
102.	Total length, min	Gage	03002
103.	Perpendicularity of rear end surface with bourrelet	Gage	03003
104.	Pitch diameter of thread	Gage	03004
105.	Minor diameter of thread, max.	Gage	03005
106.	Concentricity of pitch diameter of thread with bourrelet, including perpendicularity with nose face	Gage	03006
107.	Diameter of band seat	Gage	03007
108.	Width of band seat	Gage	03008
109.	Concentricity of band seat diameter with bourrelet	Gage	03009
110.	Distance from rear end surface to band seat	Gage	03010
111.	Diameter of base plug cavity	Gage	03011
112.	Depth of base plug cavity	Gage	03012
113.	Concentricity of base plug cavity with bourrelet	Gage	03013
114.	Body diameter between bourrelets	Gage	03014
115.	Concentricity of body diameter with bourrelets	Gage	03015
116.	Inside diameter for min depth, min.	Gage	03016
117.	Concentricity of inside diameter with bourrelet	Gage	03017
118.	Distance from nose face to smallest basic inside diameter	Gage	03018
119.	Wall thickness at bottom of cavity in specified area, min.	Gage	03019
120.	Wall thickness variation in excess of permitted amount	Gage	03020
121.	Length from nose face to larger basic outside diameter on ogive	Gage	03021
122.	Distance from rear face to basic outside diameter on boattail	Gage	03022
123.	Scoring of bandseat missing or improper	Visual	03023

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124.	Finish improper on press-fit diameter	Visual	03024
125.	Finish improper in body cavity	Visual	03025
126.	Evidence of poor workmanship on bandseat	Visual	03026
127.	Concentricity of boattail with bourrelet at indicated distance	Visual	03027
128.	Length from rear face to holes in boattail (3 places)	Gage	03028
129.	Diameter of holes in boattail (3 places)	Gage	03029
130.	Length from nose face to body undercut behind forward bourrelet	Gage	03030
131.	Length from nose face to smaller basic outside diameter	Gage	03031
132.	Distance from rear face to smaller basic outside diameter on boatail	Gage	03032
133.	Length of thread, min.	Gage	03033
Minor:	AQL 2.50 Percent		
201.	Inside diameter, max.	Gage	03034
202.	Staking notch missing, or improper	Gage	03035
203.	Finish on bourrelets improper	Visual	03036
204.	Base plug chamfer missing or improper (2 places)	Visual	03037
205.	Discoloration (due to corrosive action)	Visual	03038
206.	Evidence of poor workmanship (see 3.8)	Visual	03039

4.3.2.4 Body assembly, prior to painting (see dwg. 9245951)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Diameter of rotating band	Gage	04001
102.	Concentricity of rotating band with bourrelet	Gage	04002
103.	Length to rotating band from base face	Gage	04003
104.	Width of rotating band	Gage	04004
105.	Total length, max.	Gage	04005
106.	Band seat width not completely filled	Visual	04006
Minor:	AQL 1.00 Percent		
201.	Rotating band undercut excessive (2 places)	Gage	04007
202.	Evidence of poor workmanship (see 3.8)	Visual	04008

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4.3.2.5 Body assembly, after painting (see dwg. 9245951)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Diameter of bourrelet, max (2 places)	Gage	05001
102.	Rotating band missing or damaged	Visual	05002
Minor:	AQL 1.50 Percent		
201.	Bare spot on protective coating (ignoring light scratches)	Visual	05003
202.	Marking incomplete, incorrect or unidentifiable	Visual	05004
203.	Evidence of poor workmanship (see 3.8)	Visual	05005

4.3.2.6 Pallet, prior to assembling of cover (see dwg. 8837835)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Incorrect number of body assemblies	Visual	06001
102.	Assemblies inadequately protected from moisture	Visual	06002
103.	Assemblies incorrectly positioned (inverted)	Visual	06003
104.	Sleeve or ring assembly missing or damaged	Visual	06004
Minor:	AQL 0.65 Percent		
201.	Liner missing	Visual	06005

4.3.2.7 Pallet (see dwg. 8837835)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Hardware missing, broken or loose	Visual/Manual	07001
102.	Strapping missing, broken or loose	Visual/Manual	07002
103.	Board broken or split	Visual	07003
104.	Label missing (where applicable)	Visual	07004

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Minor:	AQL 1.50 Percent		
201.	Hardware improperly engaged	Visual/Manual	07005
202.	Strapping improperly assembled	Visual	07006
203.	Marking misleading or unidentifiable (including incorrect color code)	Visual	07007
204.	Car seal missing, unsealed or improperly positioned	Visual	07008

4.3.3 Testing

4.3.3.1 Tensile strength of body (prior to machining) (see dwg. 9245952) Major Defect, Code No. 08001.-Sample bodies shall be selected from each mill heat of steel (see 4.3.1.1) in accordance with Table I and test specimens cut therefrom. If any test specimen fails to comply with the physical properties requirements specified on the applicable drawing, that quantity of bodies represented by the sample specimen shall be rejected. The test shall be performed as specified in 4.4.1, using equipment in accordance with 4.3.4.

TABLE I

No. of Bodies in Mill Heat Lot	Sample Size
50-200	2
201-300	3
301-400	4
401-500	5
501-600	6
601-700	7
701-800	8
801-900	9
Over 900	10

If more than one heat treatment or stress relief batch is included in the mill-heat of steel, or when the heat of steel is processed in two (2) or more batch-type furnaces, the sample bodies shall be selected in such a manner as to be representative of the greatest number of batches possible within the mill heat. When heat treatment or stress relief is performed in a continuously operated, conveyor-type furnace, the samples shall be selected in a sequence that will be representative, to the maximum extent, of the entire mill heat lot in process.

4.3.3.2 Surface defects of body assembly (after phosphate coating and prior to painting) (See dwg. 9245951) Major Defect, Code No. 09001.-This inspection shall be performed 100%. Each body assembly shall be inspected visually for cracks, seams and laminations. If any body assembly contains any of these defects it shall be classed defective and removed from the lot. The examination shall be performed as specified in 4.4.2.

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4.3.3.3 Rotating band clearance (see 3.3) Major Defect, Code No. 10001.- One (1) body assembly shall be selected from each eight (8) hour's production, or fraction thereof, from each banding machine operating and subjected to this test. If any body assembly fails to comply with the applicable requirement, it shall be classed defective and the quantity of bodies banded since the last satisfactory inspection on that machine shall be rejected. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.4 Tensile strength of rotating band (see dwg. 8594851 and MIL-B-20292) Major Defect, Code No. 11001.-One (1) body assembly shall be selected from each eight (8) hour's production, or fraction thereof, from each banding machine in operation, and the band subjected to this test. If any band fails to comply with the tensile strength requirements specified, it shall be classed defective and the quantity of bodies banded on that machine since the last satisfactory inspection on that machine shall be rejected. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4. The body assemblies selected for rotating band clearance may be used for this test.

4.3.3.4.1 Retest.-A retest will be permitted at the request of the contractor. A random sample of five (5) body assemblies shall be selected from the quantity represented by each failing band and tested as specified in 4.4.4. If any rotating band fails to comply with the applicable requirements, that quantity of bodies represented by the original failing sample shall be rejected.

4.3.3.5 Bend test of rotating band (see dwg. 8594851 and MIL-B-20292) Major Defect, Code No. 12001.-One (1) body assembly shall be selected from each eight (8) hour's production, or fraction thereof, from each banding machine in operation and the band subjected to this test. If any band fails to comply with the bend test requirements specified, it shall be classed defective and the quantity of bodies banded since the last satisfactory inspection on that machine shall be rejected. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4. The body assemblies selected for rotating band clearance or tensile strength may be used for this test.

4.3.3.5.1 Retest.-A retest will be permitted at the request of the contractor. A random sample of five (5) body assemblies shall be selected from the quantity represented by each failing band and tested as specified in 4.4.5. If any rotating band fails to comply with the applicable requirements, the quantity of bodies represented by the original failing sample shall be rejected.

4.3.3.6 Salt spray test of body assembly (see dwg. 9245951) Major Defect, Code No. 13001

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4.3.3.6.1 Painted parts (Batch-type Process).-One shell body assembly shall be randomly selected from each batch and subjected to the Salt Spray Test as specified on the applicable drawing. If this part fails to comply with the specified requirement, the batch represented by this part shall be rejected. The test shall be performed as specified in 4.4.6.

4.3.3.7 Salt spray test of base plug (see dwg. 9245953) Major Defect, Code No. 14001

4.3.3.7.1 Painted parts (Batch-type process).-Fifteen (15) base plugs shall be randomly selected from each batch and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specified requirement, the batch represented by the failing part shall be rejected. The test shall be performed as specified in 4.4.7.

4.3.3.7.2 Painted parts (Continuous-type process).-Fifteen (15) base plugs shall be randomly selected from each four (4) hour's of production, or portion thereof, and subjected to the salt spray test as specified in the applicable drawing. If any part fails to comply with the specified requirement, the quantity represented by the failing part shall be rejected. The test shall be performed as specified in 4.4.7.

4.3.3.8 Hardness of base plug (see dwg. 9245953) Major Defect, Code No. 15001.-Fifteen (15) base plugs shall be randomly selected from each heat treatment batch (See Note 1) and subjected to this test. The test shall be performed as specified in 4.4.8 using equipment in accordance with 4.3.4.

NOTE 1 - A heat-treatment batch shall be defined as parts that have been heat treated at the same time in the same furnace and quench bath for each of all phases of the entire heat treatment process.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9245951 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.- The contractor shall, when required by the Government, make available all inspection and test equipment necessary for determining conformance with the contract requirements. Personnel for operating the equipment, and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

4.4.1 Tensile strength of body.-Prior to machining of the body, one (1) tension specimen conforming to either FED-STD-151, Method 211.1, Section 4, Fig. 2, Type 2, or Figures 1 and 2 of this Purchase Specification, shall be taken longitudinally from the wall of each sample body. The test specimen shall be taken in such a position that the gage length will include the section of the wall at the location of the rotating band, and in such a manner that the test specimen is from within the contours of the finished body. The test shall be conducted in accordance with Method 211.1 of FED-STD-151. Yield strength shall be determined by either the effect offset method or extension-under-load method. The limiting set shall be 0.1 percent (0.001 inch per inch of gage length). Yield point shall be determined by the divider method. Identification of each heat treated lot shall be maintained until minimum physical properties have been verified by the inspector.

4.4.2 Surface defects of body assembly.-Inspection for cracks, seams and laminations shall be performed on the interior and exterior of the body, prior to painting. Interior inspection shall be conducted by means of a light and a mirror or any other method approved by the Contracting Officer.

4.4.3 Rotating band clearance.-Prior to removal, the diameter of the finished band shall be measured on three (3) diameters, 60° apart and on two (2) points along the cylindrical portion of its length near the front and near the rear edge. This comprises six (6) diametral measurements The points of measurement shall be marked on the bands as well as the orientation of the band on the body. The band shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of twelve (12) measurements), and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat, the sum of the corresponding two (2) thicknesses of the band and subtract the result from the corresponding exterior diameter of the band previously determined. The result will indicate the total diametral clearance between the band and its seat. If the band is well seated, a negative clearance, or interference, may be indicated on some of the measurements or even on the average of all measurements. When waved band ribs or irregular band profile interfere with ease of measurement, the manufacturer is authorized to machine the outside of the band in order to obtain the adequate cylindrical surface. The diameter of the machined band must exceed the bourrelet diameter.

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4.4.4 Mechanical Properties of rotating band.-The band shall be carefully removed from the body and carefully flattened without hammering. The test specimen shall be cut along the direction of the flattened length and prepared in such a manner as to minimize any additional cold working. Tensile test specimens shall be machined to the form and dimensions specified in MIL-B-20292. The specimens shall be tested as specified in MIL-B-20292.

4.4.5 Bend test of rotating band.-The band, or section thereof, of full width and thickness, except as reduced by authorized surface cut, shall be bent cold by pressure, without hammering, in the direction of curvation of the band, until it tightly encloses 180 degrees of a rod equal in diameter to the thickness of the band test specimen (within plus or minus 0.010 inch). The band shall be observed for cracks resulting from the test.

4.4.6 Salt spray test of shell body assembly.-The shell body shall be subjected to the Salt Test as specified in FED-STD-141, Method 6061, and detailed finish specification. Examination shall be made for any evidence of corrosion. This is a destructive test. Parts so tested shall not be returned to the batch or lot.

4.4.7 Salt Spray Test of Base Plug.-The Base Plug shall be subjected to the Salt Test as specified in FED-STD-141, Method 6061, and detailed finish specification. Examination shall be made for evidence of corrosion. This is a Destructive Test. Plugs so tested shall not be returned to the batch or lot.

4.4.8 Hardness of base plug.-This test shall be conducted in accordance with the methods and procedures specified in ASTM E18-67, using the penetrator and load for the hardness scale specified on the applicable drawing. A total of three (3) surface readings shall be taken on the interior surface of each base plug. The readings shall be 120° apart at distance of one inch from the center ofthe base plug. If any reading falls outside the hardness requirement, the base plug shall be classed defective.

4.4.9 Proving ground ballistic tests (see 3.6).-The tests specified in 4.2.2.1 shall be conducted at a Government-owned proving ground. A suitable recovery area shall be prepared prior to firing any rounds. The area shall minimize the effects of impact on the body assembly and permit recovery of the body assemblies intact for examination.

All rounds shall be fired from a weapon for which they are standard, with the fuze assembly set to achieve a minimum burst height of fifteen hundred (1500) feet. Photographic equipment shall be positioned near the gun muzzle to record the flight of each round. Examination of photographs and recovered body assemblies shall be made for any evidence of failure of the metal parts in order to determine compliance of each round with the requirements of 3.6.

Those round which are to be inert-loaded may contain wax, borax, or other approved inert material in place of the explosive.

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All test results and other pertinent information shall be recorded and shall include the following (see 6.5):

a. Evidence of failure to comply with any of the requirements in 3.6.

b. Rupturing or cracking of ogive in wall area at bottom of large cavity.

c. Probable cause of defect (impact, ballistically caused, manufacturing defect, etc.)

4.4.9.1 Zone 7.-The five (5) body assemblies to be tested at Zone 7 shall be inert-loaded (may contain live primers). All rounds shall be fired at ambient temperature using the appropriate Zone 7 propelling charge increments.

4.4.9.2 Five-foot drop and Zone 7 firing.-The five (5) body assemblies shall be inert-loaded, (may contain live primers), and, with a rope sling assembled, shall have a five-foot drop test performed prior to firing. The projectile shall be dropped five feet, (lowest point of the projectile to point of impact) onto a steel plate which is solidly supported by concrete. The orientation of the projectile shall be at an axis 45 degrees from the vertical, with the base down. The rope sling shall be attached to the projectile in a manner which will prevent the projectile from tumbling after impact. The armor plate upon which impact occurs shall have a minimum thickness of 3 inches, a Brinnel hardness of at least 207 and shall be solidly supported in a horizontal plane over its entire bearing area by a minimum thickness of 24 inches of concrete. The steel plate shall have a smooth and flat surface. Upon completion of the drop test, the projectile shall be examined for evidence of separation, loosening, or fracture of any body assembly or base plug. Projectiles showing no evidence of these defects, as determined by the Proving Ground, shall then have the fuze assembled. The rounds shall be fired at ambient temperature using the appropriate Zone 7 propelling charge increments.

4.4.9.3 Zone 7X.-The ten (10) body assemblies to be tested at Zone 7X shall be HE loaded. Prior to firing, the rounds shall be temperature-conditioned for not less than 24 hours at the temperature specified ± 5°F:

Five (5) of the rounds shall be conditioned to -65°F.
Five (5) of the rounds shall be conditioned to +165°F.

The rounds shall be fired using a propelling charge adjusted to give 112 percent of the rated pressure of the propelling charge. (The rated pressure appears on the data card of each propelling charge lot.)

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-The body assemblies shall be preserved and packaged as specified on dwg. 8837835.

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5.1.1.1 Base plugs.-The base plugs shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing

5.2.1 Level C.-The body assemblies shall be packed as specified on dwg. 8837835.

5.2.1.1 Base plugs.-The base plugs shall be packed in accordance with MIL-STD-1169.

5.3 Marking.-Marking shall be in accordance with dwg. 8837835 and MIL-STD-1169.

5.4 Shipping.-When shipments from more than one lot of body assemblies or base plugs are made at one time, such as in a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.
c. Provisions for submission of first article samples, including the loading and testing of the body assemblies to be subjected to the ballistic tests specified in 4.2.2.1.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.6.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6. Action required by the contractor with respect to the use of commercial equipment is detailed in 6.3.5.

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6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4 and 6.3.5 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army designs, the contractor shall comply with, anduse these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.6.

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6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors, normally, are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.4 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.6.

6.3.6 Submission of contractor designs.-All submitted designs shall reference the applicable five-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on EL, all designs of equipment for inspection of defects classified as Critical and Major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.- Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

6.5 Results of proving ground ballistic tests.-A copy of the ballistic test results and any other requested information relating to these tests shall be forwarded to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

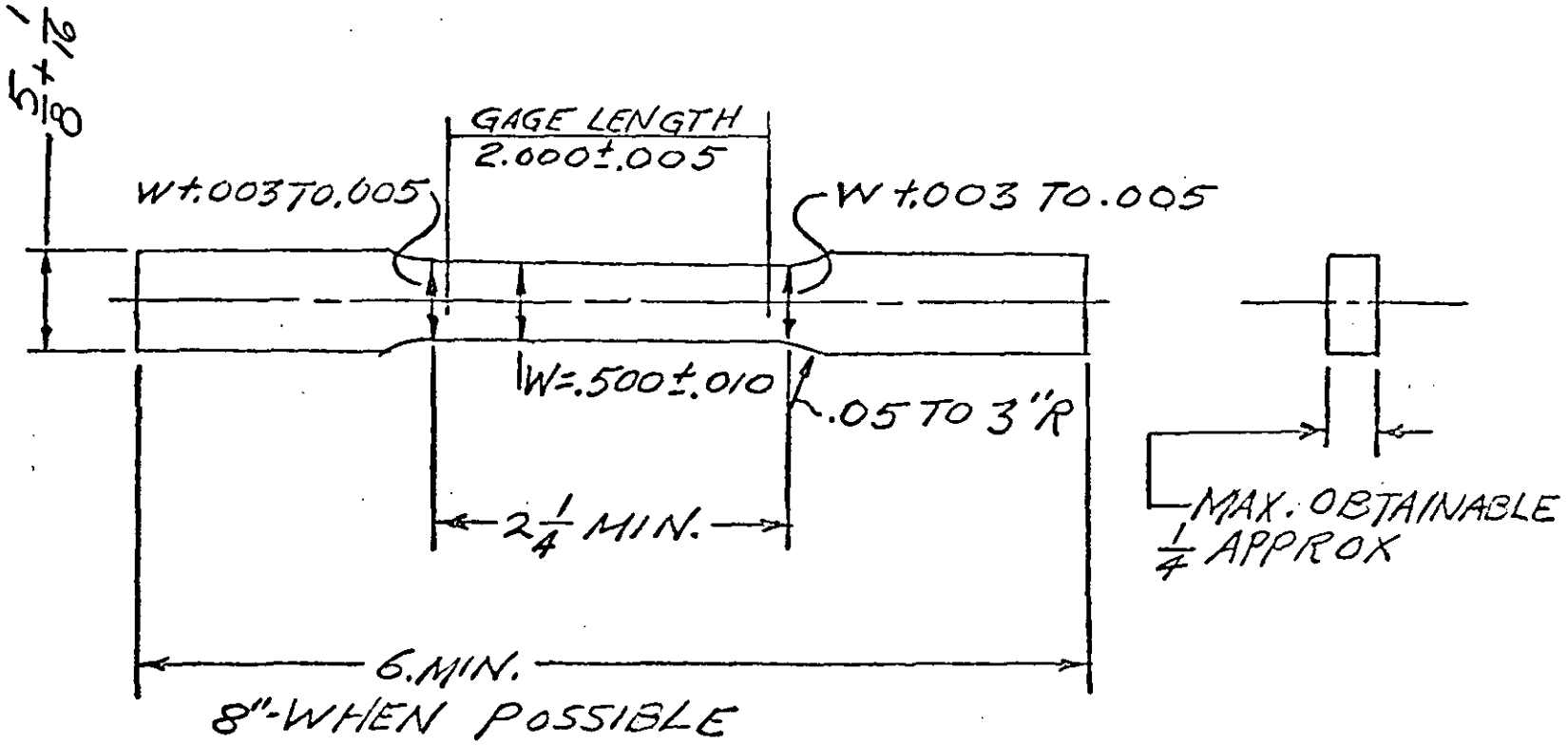
6.6 Supersession data.-This specification includes the requirements of Purchase Specification AEI-1141A dated 23 February 1968.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1315-A-681

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TENSILE TEST SPECIMEN

FIGURE 1

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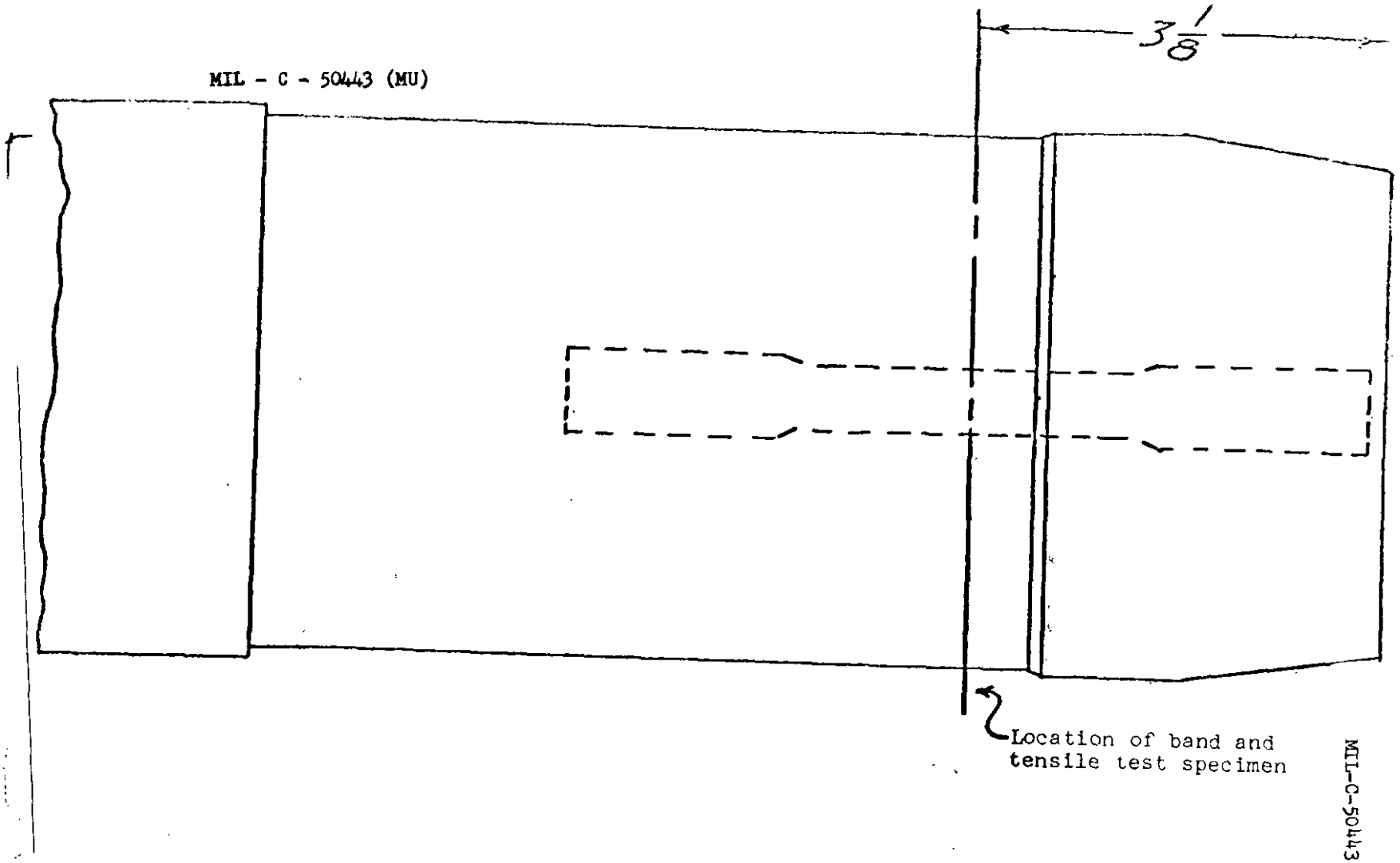


FIGURE 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION		
ORGANIZATION	CITY AND STATE	
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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